

1 REVIEW FOR EXAM I Chem 1020 Chapter 11 and 12

1. Study your notes; I have tried to organize the material putting emphasis on areas I believe are most important (and therefore will focus the exam on).
2. Read the two chapters: you are responsible for what's in them *even if I didn't cover it in class* except for these specifically excluded items: **mechanisms, polymers**.
3. Understand the definition of the terms. "Key Terms" are listed with the summaries at the end of the chapters (p. 382 & p. 417) I expect you to be able to *spell* these words.
4. Work the assigned homework problems. If you need repetition, turn your notebook to a clean page and *work the same problems again*. They should be easier and faster the second time around. If they aren't, you need to review more carefully.
5. You may expect an exam which has an objective section (multiple choice, fill in the blank), perhaps some short answer (one or two sentences), skills section (name, draw), complete the reaction, and a draw the isomers question (like Q.11.40 or 12.8 or 12.44)
6. You will **not** need a calculator. Just bring a pen or pencil. (**a pencil is highly preferred**) Model kits are permitted, although you should be aware that building the models often takes more time than you think. Model kits may not be shared.

Chapter 11

1. tetrahedral geometry around carbon for alkanes – 109.5° [C atom bonded to 4 other atoms]
2. alkanes and cycloalkanes are *saturated* (all single bonds)
3. general formula for alkanes: C_nH_{2n+2}
cycloalkanes: C_nH_{2n} (can have geometric (*cis*-, *trans*-) isomers, but this tends to be more important for alkenes)
4. nomenclature: name or draw - there will a lot of points devoted to this.
5. conformational isomers – recognize and understand that they are formed by free rotation around single bonds, also called "conformers" and are not really true isomers
6. isomers : *e.g.*, "draw all the isomers of..." Problem 11.38, 11.40, 11.49, and 12.17
7. Reactions: (1) complete combustion, p. 382 Reaction 1.
the only reaction you will need to be able to balance is combustion

Chapter 12

1. geometry: alkenes – trigonal planar – 120° [C atom bonded to 3 other atoms]
alkynes – linear – 180° [C atom bonded to 2 other atoms]
2. alkenes, alkynes, and aromatics are all *unsaturated* (contain at least one multiple bond)
3. general formula for alkenes: C_nH_{2n} and for alkynes: C_nH_{2n-2}
4. nomenclature: name or draw- a lot of points here
5. geometric isomers *cis*-, *trans*- geometry for alkenes (**be sure to show geometry if known**)
6. Reactions: Addition Reactions * these reactions follow Markovnikov's Rule
 - (1) hydrogenation
 - (2) halogenation
 - (3) hydration*
 - (4) hydrohalogenation*
7. Aromatics – general composition, compounds of benzene (C_6H_6), nomenclature (*ortho*, *meta*, *para* for disubstituted benzenes)
8. Physical properties of hydrocarbons (low m.p., low b.p., low density, nonpolar, insoluble in water, soluble in nonpolar solvents, etc.)